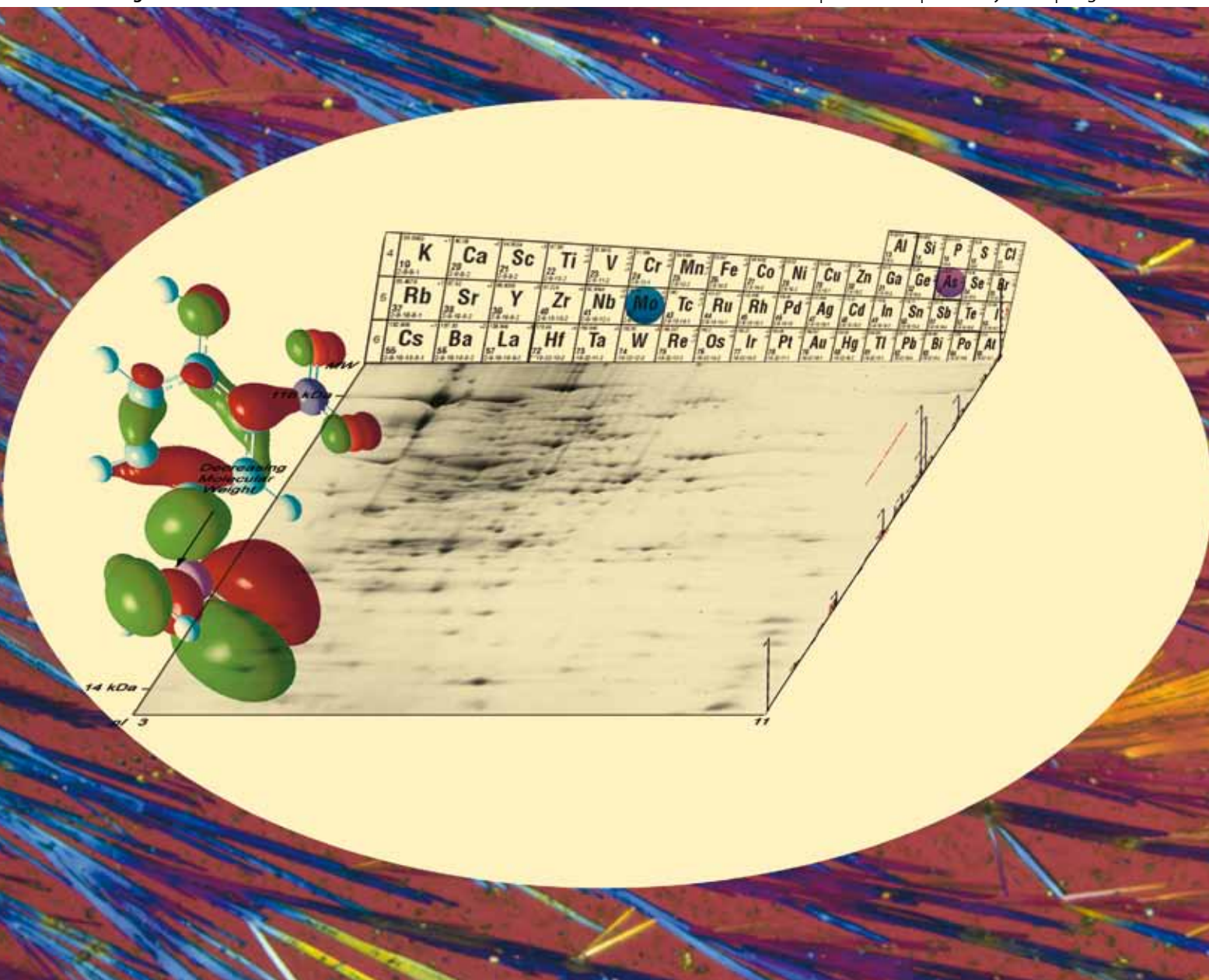


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A proteome investigation of roxarsone degradation by *Alkaliphilus oremlandii* strain OhILAs

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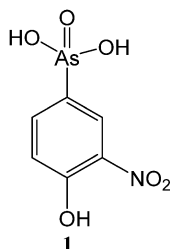
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Clostridial species predominate in both chicken gastrointestinal tract as well as litter where the organoarsenical roxarsone (3-nitro 4-hydroxybenzenearsonic acid) is anaerobically transformed releasing the more recognized toxic inorganic arsenic. 2D-gel electrophoresis and mass spectrometry were used to evaluate the changes in protein expression of *Alkaliphilus oremlandii* in response to different growth conditions (e.g., terminal electron acceptors) in order to explore the mechanism of microbial biotransformation of roxarsone. Aldehyde ferredoxin oxidoreductase, the enzyme that belongs to the xanthine oxidase family of molybdoenzymes was significantly overexpressed in the presence of roxarsone suggesting a role in the anaerobic metabolism of this substituted nitrophenol.

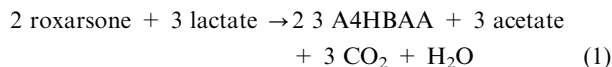
Introduction

Roxarsone, 3-nitro 4-hydroxybenzenearsonic acid (structure 1), has seen widespread use as a feed additive in concentrated agricultural farming operations (CAFOs), particularly the poultry industry. It is used to control coccidiosis, a disease caused by a protozoal parasite, with the added benefit of enhanced growth. The majority of this organoarsenical is excreted in the manure, and is not retained in the chicken tissue. Since 1955 the annual production of broilers in the USA has increased from one billion to over 9 billion in 2008.¹ In many cases, roxarsone was fed to these animals and the litter may contain from 15 to 48 mg kg⁻¹ of arsenic.^{2–5} While organoarsenicals are thought to be innocuous, recent evidence indicates that some of the organoarsenicals may be more proangiogenic than the more common inorganic arsenicals.⁶ For example, three dimensional cell based assays have shown that roxarsone may have more angiogenic potential than arsenite.⁷



Organoarsenicals are thought to be stable and not degrade in the environment; however, it is now known that roxarsone can be degraded to inorganic arsenic due to microbial activity^{8,9} potentially posing an environmental concern.¹⁰ We have demonstrated^{11,12} that roxarsone can be converted to

inorganic arsenic by anaerobic bacteria when grown with lactate as an electron donor. The measurable end products were acetate, 3-amino-4-hydroxy-benzenearsonic acid (3A4HBAA), and arsenate, all of which have been quantitated for mass balance. The products are the same whether the roxarsone is degraded by microbes present in the chicken litter or in a pure culture of *Alkaliphilus oremlandii*. The overall process that consumes lactate producing acetate is shown in eqn (1).



Interestingly, both 3A4BAA and arsenate are produced simultaneously suggesting parallel pathways may be operative. Genome analysis of *A. oremlandii* indicates the presence of many potential candidates, however, which proteins are involved in the biotransformation of roxarsone remains unclear. Even at a more fundamental level whether specific proteins are expressed by *A. oremlandii* under different growth conditions is not known. Herein we report proteomic results in an effort to identify proteins that are differentially expressed in cells grown on roxarsone as compared to those grown on lactate alone. The latter cultures (i.e., cells grown on lactate alone) provided a means to identify constitutively as well as differentially expressed proteins. Our results indicate that one protein in particular, aldehyde ferredoxin oxidoreductase, a molybdopterin-containing protein, is expressed more in the roxarsone grown cells.

Materials and methods

Cell growth. *A. oremlandii* strain OhILAs is a low G + C gram positive bacterium isolated from anoxic sediments from the Ohio River (Pittsburgh, PA, USA) that is capable of dissimilatory arsenate reduction and roxarsone transformation.¹² Concentrations of 2.5 and 5 mM roxarsone inhibited growth of the organism, while the growth yields were enhanced with 0.5 and 1 mM roxarsone with maximal growth observed with 1 mM roxarsone.^{11,12} For the current set of experiments, cells

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were grown anaerobically in minimal media with either 1 mM roxarsone with 10 mM lactate or 10 mM lactate alone as described.¹¹ The cells were harvested after 72 h of growth, and lysed by French pressure cell.

Protein sample preparation and 2-D electrophoresis (2-DE)

The cell lysates were precipitated with trichloroacetic acid (TCA) at a final concentration of 10% on ice for 60 min. The protein pellet obtained by subsequent centrifugation was washed with acetone in order to remove residual TCA. Immobililine DryStrips (18 cm, pH 3-11 NL, GE Healthcare) were rehydrated in 350 μ L rehydration solution containing 250 μ g solubilized proteins for IEF. The rehydration solution consisted of 7 M urea, 2 M thiourea, 4% (w/v) CHAPS, 40 mM DTT and 2% (v/v) IPG buffer and also contained a trace of bromophenol blue as tracking dye. Rehydration was allowed to proceed overnight to ensure maximal diffusion (attachment) of the proteins to the strips. Isoelectric focusing was carried out on an Ettan IPGphor3 unit (GE Healthcare) at 300 V for 2 h, 500 V for 1 h and 8 kV, to achieve a total of 70 kVh at 20 °C. Following IEF, the Drystrip cover fluid was poured off and the strips were equilibrated twice for 15 min each time with gentle shaking, first in 30% (v/v) glycerol, 6 M urea, 2% (w/v) SDS, 50 mM Tris-Cl, pH 8.8, 1% (w/v) DTT and a trace of bromophenol blue as tracking dye and the second time in this same solution, except that DTT was replaced by 2.5% (w/v) iodoacetamide. After this equilibration, each IEF strip was drained on a filter paper and immersed in SDS running buffer (25 mM Tris base, 0.192 M glycine and 0.1% SDS) before being sealed at the top of the PAGE gel with 1% agarose in SDS running buffer. The second-dimensional electrophoresis was run on 12.5% SDS-PAGE using an Ettan Dalt six electrophoresis unit (GE Healthcare) at 4 °C, 15 mA for 60 min followed by 30 mA for the remaining run. Electrophoresis was performed until the tracking dye reached the anodic end of the gels.¹³ Three replicate gels were run. For proteomic analysis, protein spots were visualized employing Coomassie Brilliant Blue G-250, according to Neuhoff *et al.*¹⁴ The stained gels were scanned by using a GS-800 densitometer (Bio-Rad). Image analysis was carried out by using the NIH Image J software. The most intense spots were analyzed first followed by less intense ones until reliable mass spectra could be obtained.

Matrix-assisted laser/desorption ionization-time of flight mass spectrometry (MALDI-TOF-MS)

Protein spots were excised from the gel using a pipette, and transferred into 1.5 mL Eppendorf tubes. In-gel trypsin (Promega, USA) digestion and sample preparation for MALDI-TOF analysis was carried out by the procedure of Shevchenko *et al.*¹⁵ After the overnight digestion, the samples were concentrated and desalted by ZipTip (Millipore) prior to applying on the sample plate. α -Cyano-4-hydroxycinnamic acid (Fluka, Switzerland) was recrystallized and used to prepare the matrix solution by dissolution at a concentration of 10 mg mL⁻¹ in 70% acetonitrile/0.1% TFA. 0.5 μ L peptide solution was deposited onto the MALDI target, rapidly followed

by 0.5 μ L of this matrix solution. The MALDI-TOF analysis was performed on a Voyager-DE STR MALDI-TOF mass spectrometer from Applied Biosystems (Foster City, CA) in the reflector mode. External mass calibration was achieved utilizing calibration mixture 2 of the Sequazyme peptide mass standard kit (Perseptive Biosystems). The proteins were identified by searching a bacterial subset in the NCBI database using the Mascot program (www.matrixscience.com). The search parameters were set up as follows: enzyme was trypsin, the number of missed cleavage site was allowed up to 1, the variable modification was oxidation of methionine, the mass tolerance of precursor ions and fragments was 0.5 Da. The estimated molecular weight and pI of the candidate protein were in agreement with the assignment.

Results

Cultures of *A. oremlandii*, regardless of growth medium content, grew to maximum cell densities by 72 h. During this time, the yellow color of the medium that contained roxarsone disappeared, indicating its transformation.¹¹

Proteomic analysis of cell lysates employing 2-D Gel Electrophoresis (2-DE) and MALDI-TOF Mass Spectrometry (MALDI-TOF MS)

In order to analyze the changes in protein profile affected by roxarsone, 2-DE was employed to compare the protein expressions of the cells grown on different substrates. In this case, cells grown on lactate only serve as a control for differential expression. Isoelectric focusing was conducted in the pH range 3 to 11, which provided sufficient resolution for subsequent SDS-PAGE separation in the second dimension. From the stained gels, samples were excised for tryptic digestion for mass spectrometric analysis by MALDI-TOF on the abundant proteins. This approach resulted in the definitive identification of 70 proteins that corresponded to predicted protein-encoding gene models in the annotated genome of *A. oremlandii* (Table 1). While several spots in the gel were clearly different in their intensity, quantitative image analysis was inconsistent and thus was not followed further. Protein assignments were made by MALDI-TOF analysis, searching a subset in the NCBI database using the Mascot program. For protein fractions from both cells grown on lactate and cells grown on lactate with roxarsone, three biological triplicates were used for 2-DE analyses and many proteins were identified multiple times from multiple gels. The identified proteins (with names as annotated), with their GI accession number, molecular weight (MW), and isoelectric point (pI) are listed in Table 1. Representative 2D gels (stained with colloidal Coomassie Blue) showing the presence of a large number of proteins are shown in Fig. 1. The number and relative intensity of the protein spots appear to be similar whether the cells were grown on lactate alone (Fig. 1A) or lactate and roxarsone (Fig. 1B). Nevertheless, several spots were more intense in the roxarsone grown cells, in particular aldehyde ferredoxin oxidoreductase (AOR) (Fig. 1 inset).

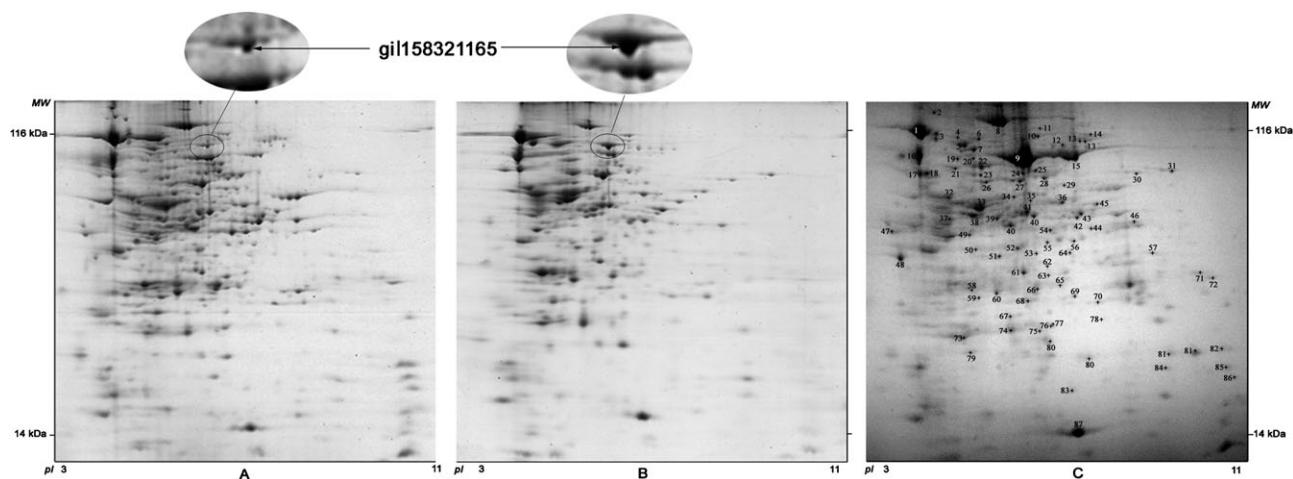


Fig. 1 Representative 2D gels (12.5% acrylamide, 3–11 pH range) of protein fractions from cells of *A. oremlandii* grown on lactate only (A) and lactate with roxarsone (B). The protein spots that were excised, in-gel digested with trypsin, and analyzed by MALDI-TOF-MS are shown in (C). The numbers correspond to the list of proteins in Table 1. The inset is a higher magnification of the spots identified as AOR.

Discussion

The mass spectrometric analysis revealed the identity of 70 proteins. While the majority of the proteins detected have known function, four proteins (Clos_2624, Clos_2626, Clos_0327, Clos_2214) were annotated as hypothetical proteins in the *A. oremlandii* genome (<http://genome.ornl.gov/microbial/clos/>). While the functions of these proteins are yet to be understood, they are nevertheless expressed under the growth conditions used in this study, and therefore are no longer hypothetical. In addition, the protein designated as molybdopterin oxidoreductase has now been confirmed to be respiratory arsenate reductase (Clos_1096, gi|158320131).¹⁶ Proteomic studies indicate that one of the major proteins is S-layer or surface layer protein, which is a common structural component for bacterial cells. Consistent with the cell growth this protein is prominently expressed both in lactate grown and roxarsone with lactate grown cells. Several proteins such as glycine dehydrogenase and Glu/Leu/Phe/Val dehydrogenase that are involved in the amino acid metabolism were identified. Similarly, proteins involved in transcription and translation were also identified, as well as those (e.g., Inosine-5'-monophosphate dehydrogenase) involved in nucleotide biosynthesis. While many of these proteins are of interest, we focus here on those that may be involved in organoarsenical biotransformation.

Previous studies demonstrated that under the conditions the cells were grown, two major products were detected.¹¹ Nearly 66% of the roxarsone was converted to 3-amino-4-hydroxy-benzenearsonic acid (3A4HBAA) through reduction of the nitro-group. The remaining roxarsone was converted to arsenate, which resulted from either direct ring cleavage or complete conversion of the aromatic ring into a C1 compound. We hypothesized the latter mechanism may be operating.¹¹ At the same time lactate is consumed and acetate is produced.

In principle, roxarsone can accept reducing equivalents that are generated during fermentation (with ATP generated by substrate level phosphorylation of acetyl-CoA), or a terminal electron acceptor linked to oxidative phosphorylation.

The genome of *A. oremlandii* contains genes that encode for proteins required for both the fermentation of lactate and glycolysis. Consistent with this, pyruvate kinase that is involved in the final step in glycolysis in the conversion of phosphoenolpyruvate to pyruvate, with concomitant phosphorylation of ADP to ATP, was identified by MALDI-TOF MS (Table 1). In the acrylate pathway, that generates acetate, enzymes such as lactate dehydrogenase, pyruvate-ferredoxin oxidoreductase, phosphotransacetylase, acetate kinase are involved. Many of these enzymes are found to be present in the genome, but of these, only pyruvate flavodoxin/ferredoxin oxidoreductase (Clos_0237, gi|158319289) has been detected in the roxarsone grown cells. Thus, acrylate pathway may be operative regardless whether roxarsone is present or not in the growth media.

The reduction of nitrophenols can proceed under aerobic as well as anaerobic conditions and can be carried out by a range of organisms¹⁷ either by oxidation or by reduction in the presence of an exogenous carbon source to amino aromatics. The reduction of nitro groups on aromatic compounds by *Clostridia* is well known.¹⁸ For example, *C. celerescence* is capable of using hexahydro-1,3,5-trinitro-1,3,5-triazine (RDX) as a nitrogen source.¹⁹ Furthermore, the reduction of the nitro group has been shown to be dependent on the availability of reducing equivalents.²⁰ Genomic analysis of *A. oremlandii* revealed the presence of two homologues of pentaerythritol tetranitrate reductase (NemA), several nitro-reductases, carbon monoxide dehydrogenase, and hydrogenase, all of which have been implicated in the reduction of nitro groups on aromatic compounds. Thus, the reduction of the nitrogroup in roxarsone, in principle, can be carried out by any of these enzymes where the reducing equivalents could be generated from the fermentation of lactate. We have identified five nitroreductases in the genome of *A. oremlandii*. The five homologues differ in amino acid sequence, molecular mass, and pI and with varying operon structures. Indeed sequence alignment of the nitroreductases shown in Fig. 2 indicates very little homology. Of those, we have detected the largest one

Table 1 Proteins identified from 2-D PAGE (first 12 are metalloproteins)

Spot ^a	Accession No ^b	Locus ^c	Protein name ^d	MW ^e	pI ^f	Coverage ^g
8	gi 158319289	Clos_0237	Pyruvate flavodoxin/ferredoxin oxidoreductase domain protein	129350	5.39	49
9	gi 158321165	Clos_2140	Aldehyde ferredoxin oxidoreductase	78249	5.58	65
13	gi 158320131	Clos_1096	Molybdopterine oxidoreductase	96667	6.32	63
14	gi 158319915	Clos_0879	Aldehyde oxidase and xanthine dehydrogenase molybdopterine binding	86888	6.22	32
15	gi 158321168	Clos_2143	Aldehyde ferredoxin oxidoreductase	77646	6.1	61
25	gi 158320201	Clos_1166	Oligoendopeptidase F	70655	5.72	40
39	gi 158320977	Clos_1950	Flavodoxin/nitric oxide synthase	44692	5.3	70
65	gi 158321150	Clos_2125	2-oxoglutarate ferredoxin oxidoreductase beta subunit	31697	5.88	75
67	gi 158320077	Clos_1041	Pyruvate ferredoxin/flavodoxin oxidoreductase	21142	5.27	86
74	gi 158321353	Clos_2331	Rubryerythrin	22901	5.63	81
80	gi 158321166	Clos_2141	4Fe-4S ferredoxin iron-sulfur binding domain protein	25858	7.35	49
81	gi 158321169	Clos_2144	FeS Protein	25577	8.29	56
87	gi 158319291	Clos_0239	Desulfoferredoxin ferrous iron-binding region	13771	5.87	95
1	gi 158321647	Clos_2627	S-layer domain protein	101935	4.79	55
2	gi 158319535	Clos_0484	DNA-directed RNA polymerase, beta subunit	139338	4.8	26
3	gi 158321644	Clos_2624	S-layer protein	82536	4.9	63
4	gi 158320998	Clos_1971	Phenylalanyl-tRNA synthetase, beta subunit	89397	5.11	52
5	gi 158319540	Clos_0489	Translation elongation factor G	76607	5.07	81
6	gi 158320724	Clos_1695	Cobalamin B12-binding domain protein	82413	5.18	44
7	gi 158320991	Clos_1964	S-layer domain protein	86229	5.28	28
10	gi 158321286	Clos_2264	Ig-like, group 2	102488	5.84	53
12	gi 158321863	Clos_2845	Peptidoglycan-binding domain 1 protein	79856	8.07	42
16	gi 158321646	Clos_2626	Hypothetical protein Clos_2626	73325	4.83	58
17	gi 158321468	Clos_2447	Chaperonin GroEL	57624	4.76	63
19	gi 158319637	Clos_0586	Pyruvate kinase	62886	5.06	70
20	gi 158320738	Clos_1709	Aspartyl-tRNA synthetase	67435	5.16	59
21	gi 158321621	Clos_2601	S-layer domain protein	69204	5.2	34
22	gi 158320145	Clos_1110	Arsenite-activated ATPase ArsA	64704	5.24	64
23	gi 158319487	Clos_0436	Methylmalonyl-CoA mutase, large subunit	62272	5.26	52
24	gi 158321392	Clos_2371	RNA-metabolising metallo-beta-lactamase	62384	5.53	30
26	gi 158319518	Clos_0467	Glutamyl-tRNA synthetase	56814	5.23	52
27	gi 158319995	Clos_0959	Sarcosine reductase	55790	5.52	64
28	gi 158321152	Clos_2127	Formate-tetrahydrofolate ligase	60122	5.76	50
29	gi 158319446	Clos_0395	Acetyl-CoA carboxylase, biotin carboxylase	51256	5.88	72
30	gi 158319584	Clos_0533	Inosine-5'-monophosphate dehydrogenase	52507	7.16	37
31	gi 158320253	Clos_1218	DEAD/DEAH box helicase domain protein	59230	8.4	66
32	gi 158319889	Clos_0851	Aminotransferase class I and II	48776	4.99	72
33	gi 158319527	Clos_0476	Translation elongation factor Tu	44098	5.22	76
36	gi 158320729	Clos_1700	Glycine hydroxymethyltransferase	45439	5.9	46
37	gi 158320624	Clos_1592	Phosphopentomutase	42818	5.03	76
38	gi 158321514	Clos_2493	FAD-dependent pyridine nucleotide-disulfide oxidoreductase	47479	5.3	63
40	gi 158321641	Clos_2621	Outer membrane protein-like protein	45956	5.83	68
41	gi 158321480	Clos_2459	Glu/Leu/Phe/Val dehydrogenase	45250	5.56	73
42	gi 158319685	Clos_0636	Acyl-CoA dehydrogenase domain protein	41315	6.06	79
43	gi 158320699	Clos_1670	Cysteine desulfurase	43625	6.07	45
44	gi 158320484	Clos_1451	Fatty acid/phospholipid synthesis protein PlsX	36404	6.07	86
45	gi 158319945	Clos_0909	Pyridoxal phosphate-dependent acyltransferase	43367	6.23	71
46	gi 158320194	Clos_1159	FMN-dependent alpha-hydroxy acid dehydrogenase	36349	6.96	81
47	gi 158319997	Clos_0961	Basic membrane lipoprotein	38677	4.46	59
48	gi 158319379	Clos_0327	Hypothetical protein Clos_0327	44008	4.37	52
49	gi 158319078	Clos_0019	Lipoyltransferase and lipoyl-protein ligase	38307	5.28	62
51	gi 158319585	Clos_0534	GMP synthase, large subunit	36058	5.36	80
52	gi 158319301	Clos_0249	Methylenetetrahydrofolate dehydrogenase (NADP(+))	31199	5.43	63
54	gi 158319458	Clos_0407	Branched-chain amino acid aminotransferase	37699	5.74	39
55	gi 158319760	Clos_0712	D-isomer specific 2-hydroxyacid dehydrogenase NAD-binding	36170	5.71	51
56	gi 158321288	Clos_2266	Transketolase central region	33448	5.94	83
58	gi 158320546	Clos_1514	Translation elongation factor Ts	24318	5.2	67
60	gi 158319309	Clos_0257	Short-chain dehydrogenase/reductase SDR	26077	5.27	43
61	gi 158319300	Clos_0248	Nitroreductase	30984	5.46	87
66	gi 158319887	Clos_0849	Short-chain dehydrogenase/reductase SDR	27513	5.64	60
68	gi 158321147	Clos_2122	CBS domain containing protein	22284	5.6	52
69	gi 158321130	Clos_2105	HAD-superfamily subfamily IB hydrolase	28373	6	68
70	gi 158319749	Clos_0701	Stress protein	21929	6.08	72
73	gi 158319724	Clos_0675	Redoxin domain protein	19047	5.11	88
75	gi 158320386	Clos_1352	ATP-cobalamin adenosyltransferase	20342	5.64	81
76	gi 158319190	Clos_0134	Isochorismatase hydrolase	20665	5.79	65
77	gi 158320393	Clos_1360	Propanediol utilization protein	20972	5.75	49
78	gi 158319530	Clos_0479	NusG antitermination factor	19838	6.36	61
83	gi 158320813	Clos_1784	NUDIX hydrolase	17427	5.93	97
85	gi 158319578	Clos_0527	Ribosomal protein L13	16124	9.7	74

^a Spot numbers corresponding to those shown in Fig. 1. ^b Accession number from the NCBI genome sequence database for the gene corresponding to the *A. oremlandii*. ^c Locus tag number of the protein. ^d The protein annotation. ^e The predicted molecular mass of the identified protein. ^f The predicted pI of the identified protein. ^g MS sequence coverage.

(Clos_0248, gi|158319300) indicating that at least this protein is expressed. Future functional assays will reveal whether this protein can reduce the nitro group in roxarsone. From the analysis of the gel images, it appears that this nitroreductase is present in both conditions, thus being expressed constitutively.

We hypothesized that the release of inorganic arsenic from roxarsone follows complete degradation of the aromatic ring. Aerobic degradation of nitroaromatics that serve as a source of carbon and nitrogen are known to proceed through complete conversion to inorganic species. Often these reactions are carried out by monooxygenases or dioxygenases. For example, monooxygenase from *Pseudomonas sp.*, has been demonstrated to convert 4-nitrophenol to hydroquinone, with the release of nitrite at the expense of two moles of NADPH per mole of 4-nitrophenol.²¹ The hydroquinone formed served as a substrate for ring fission forming γ -hydroxy muconic semialdehyde, which is further oxidized to maleyl acetic acid and then to β -ketoadipic acid.²¹ Alternatively the degradation can proceed *via* hydroxylation of the ring followed by ring

fission to maleyl acetic acid and finally to β -ketoadipic acid.²² Both processes proceed aerobically and suggest an alternative pathway may be operative in the present case. Boll and others have proposed that the anaerobic degradation of the benzene ring proceeds *via* a benzyl CoA reductase pathway where the CoA activates the ring followed by reduction of the ring, which is subsequently degraded to CO₂.^{23–26} Interestingly, aldehyde ferredoxin oxidoreductase (AOR), a pterin containing molybdoprotein, was found to be expressed at a higher level in cells grown in roxarsone (see Fig. 1, inset). The genome of *A. oremlandii* contains two homologues of AOR and both copies are found to be expressed at higher level, but to different extents, in roxarsone grown cells. AOR is a hydroxylase that belongs to the xanthine oxidase family of molybdenum enzymes that can hydroxylate an organic substrate. This enzyme is in the same family as benzoyl CoA reductase. Based on the 2-DE data it is provocative to hypothesize that AOR may be involved in the degradation of roxarsone. Whether roxarsone can function as a direct

gi 158319693 ref YP_001512200.	-----	
gi 158321769 ref YP_001514276.	-----	
gi 158319788 ref YP_001512295.	MNRNSTIKIDVEKICGCLCKDCPLHVLVLQNGKATMMAEKCLECGHCV	50
gi 158320190 ref YP_001512697.	-----MSKGIGREFMEKTQYRYLEESDQNKGLPQPPLVPTDS	38
gi 158319300 ref YP_001511807.	----- MTFINLVENR KSVREYQKLLGAKEMKALEK ALEEIDVL	39
gi 158319693 ref YP_001512200.	-----MSYLDIFYKRHSVRKFKNEE	20
gi 158321769 ref YP_001514276.	-----MKEIFERRSIRKYKGKE	17
gi 158319788 ref YP_001512295.	AICPKDAFTMRGYDMQEVKTYDKNI FGIDEDILNT IQFRS VRHYKDRQ	100
gi 158320190 ref YP_001512697.	NVALINLTPVKDIN-----VQRIDLREAIERRKSLRKYADTS	75
gi 158319300 ref YP_001511807.	IKNVHVDFAFIEDG ----- WGKEILRGRAGYVGNPVLAPHYIA	78
gi 158319693 ref YP_001512200.	VPLGDIQEIITAATHAPSGKNLQNHVVTNKEIIRGIADTIVKKAKAI	70
gi 158321769 ref YP_001514276.	ISQDYIQKILRAAMYAPSAGNEQFWHFVVIKDRQKLDEIAS-----	58
gi 158319788 ref YP_001512295.	VEKEVIKKIIEAGRETPTGSNKQIRYIQAQESIPLNEALKTFR----	146
gi 158320190 ref YP_001512697.	LTMEELSYLLWCTQGVKQIVSRPATIRTVPSAGARHPFETYLLINN----	121
gi 158319300 ref YP_001511807.	LISEEKEGYLLNSSYILEQIVLKAVELD IASCWISVENDAAGLRDLG --	126
gi 158319693 ref YP_001512200.	TASINDEALKTGFLSRIRHATLFTNAPVTILIFAGHYPTGLDILEAGGA	120
gi 158321769 ref YP_001514276.	-----FHP---HGQMLKEAPLAIIPCADQSNLK-----	83
gi 158319788 ref YP_001512295.	-----KLKRLLEIVGKFKVQLPYDVSGYQLEPGFFFHG-----	178
gi 158320190 ref YP_001512697.	-----VTGLEPGLYRYIALSHQLMVVNLEAEIDDK-----	151
gi 158319300 ref YP_001511807.	----- IDKPGEIVAMIALGYPKTHIPYTKKSGSSRIGVEDFVFKG ----	166
gi 158319693 ref YP_001512200.	TREEIHRLAAASPGIONIGAAVQTLMLAAANMGYG-----TCWM	159
gi 158321769 ref YP_001514276.	-----YEGLFWIQDMSACIQNI LLQSKALGLG-----TCWC	114
gi 158319788 ref YP_001512295.	---APTVLVVISDDVDASLASMSMELMAEAMGLG-----TLYV	214
gi 158320190 ref YP_001512697.	---IMKACLEQEFIKNSAVTFIWTVDMYRTGWRYG-----ERGY	187
gi 158319300 ref YP_001511807.	-- DWGNVPTHEELEQM GIEQALHSIRFAPSWANLQPWKL IIDKDQIILAV	214
gi 158319693 ref YP_001512200.	AGANFAGKEIAD----FIQFEKEGYHLVAITPLGIPEESELSSPARKPL	204
gi 158321769 ref YP_001514276.	G--VYPREELMEGM--AKLVHLPAHITPVAVIAVGPAPAE-EREVRERYNP	159
gi 158319788 ref YP_001512295.	GFFTVAAKHNKKIR--KELGLTKK-EKVVTCLAMGYADVQYMRTPRKKA	261
gi 158320190 ref YP_001512697.	RYIHLDA GHVCON---LYLSAQSIDSGVCAIAAFDDEEINSI IGLDGENH	234
gi 158319300 ref YP_001511807.	GGKEVSKK HLMLDAGIMMLYMKAFSSVGITADWSIYKEELDGSLSYHI	264
gi 158319693 ref YP_001512200.	EEIMTIL-----	211
gi 158321769 ref YP_001514276.	DRIHNESW----	167
gi 158319788 ref YP_001512295.	EIEWR-----	266
gi 158320190 ref YP_001512697.	FTVYVATVGKVL	246
gi 158319300 ref YP_001511807.	PSEXTAVGILHI	276

Fig. 2 Sequence alignment by Clustal 2.0.11 of five homologues of nitroreductase. Identified homologue Clos_0248, gi|158319300, in bold red are the peptide fragments identified by MALDI-TOF (79% coverage).

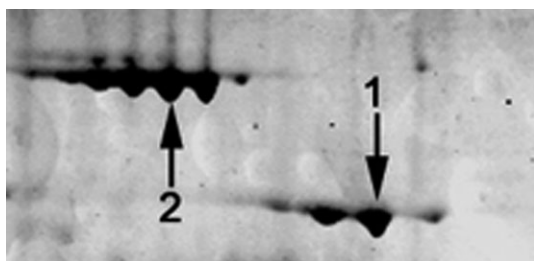


Fig. 3 Separation of aldehyde ferredoxine oxidoreductase (1) and pyruvate flavodoxin/ferredoxin oxidoreductase (2) employing a pH range 4–7 strip. The beating pattern is indicative post-translational modifications. These two proteins correspond to spot numbers 9 and 8, respectively in Fig. 1.

substrate to AOR or its degradation product serves as a substrate remains to be examined. The identification of the AOR is not limited to only one particular spot in the gel, rather peptide sequences obtained from the mass spectral data from several spots matches with the AOR. These spots do not differ in molecular mass but rather differ in their pI. Thus, we suggest that post translational modification of AOR may be taking place (Fig. 3). In addition, genomic data revealed the presence of two different homologues of AOR (Clos_2140, gi|158321165, and Clos_2143, gi|158321168) and both were detected by mass spectral analysis. From the 2-DE, it appears that one of them (Clos_2140, gi|158321165) is expressed more than the other. Thus, both enzymes are expressed under similar regulatory conditions. Unlike the AOR from thermophilic organisms such as *Pyrococcus furiosus* which harbors W-containing AOR,²⁷ AOR from *A. oremlandii* is associated with an iron sulfur cluster protein. Both copies of the AOR have this iron-sulfur cluster protein and they (Clos_2141, gi|158321166, and Clos_2144, gi|158321169) have been detected by mass spectrometry. Thus, the AOR from *A. oremlandii* is compositionally different from that of the *P. furiosus*.

Another important question is whether roxarsone induces the arsenic resistance and respiratory genes. The resistance (*ars*) and respiratory (*arr*) operons are part of a large gene cluster and are separated by molybdenum cofactor biosynthetic genes. Recently, we have described the respiratory arsenate reductase functions as a bidirectional enzyme in certain bacteria including *A. oremlandii*.¹⁶ We have identified ArrA, which along with a smaller subunit ArrB make the functional arsenate reductase, in both lactate and roxarsone grown cells. Therefore, we conclude that respiratory arsenate reductase *A. oremlandii* is constitutively expressed.

Summary

While bacterial degradation of roxarsone has been previously reported,^{9,11,12} the present study constitutes the first example of a proteomic investigation aimed at understanding the molecular mechanism of the process. We have utilized 2D gel electrophoresis coupled with mass spectrometry for identifying candidate proteins that might be involved in the roxarsone biotransformation. To this end, we have identified aldehyde oxidoreductase, nitroreductase, respiratory arsenate reductase, and arsenite activated ATPase, among others.

We have also identified proteins that are important for glycolysis and amino acid metabolism. This investigation also confirmed the existence of four different proteins that were annotated as ‘hypothetical’. It further demonstrates the utility of proteomics in the elucidation of new and novel biodegradative pathways.

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